C. Increase and Copy

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: binary search, constructive algorithms, math | https://codeforces.com/problemset/problem/1426/C

Initially, you have the array a consisting of one element 1 (a = [1]).

In one move, you can do one of the following things:

- Increase some (**single**) element of a by 1 (choose some i from 1 to the current length of a and increase a_i by one);
- Append the copy of some (**single**) element of a to the end of the array (choose some i from 1 to the current length of a and append a_i to the end of the array).

For example, consider the sequence of five moves:

1. You take the first element a_1, append its copy to the end of the array and get a = [1, 1].
2. You take the first element a_1, increase it by 1 and get a = [2, 1].
3. You take the second element a_2, append its copy to the end of the array and get a = [2, 1, 1].
4. You take the first element a_1, append its copy to the end of the array and get a = [2, 1, 1, 2].
5. You take the fourth element a_4, increase it by 1 and get a = [2, 1, 1, 3].

Your task is to find the **minimum** number of moves required to obtain the array with the sum at least n.

You have to answer t independent test cases.

Input

The first line of the input contains one integer t ($1 \le t \le 1000$) — the number of test cases. Then t test cases follow.

The only line of the test case contains one integer n ($1 \le n \le 10^9$) — the lower bound on the sum of the array.

Output

For each test case, print the answer: the **minimum** number of moves required to obtain the array with the sum at least n.

Example

Input
5 1 5 42 1337 1000000000

Output
0 3 11 72 63244